

2026 年（第十三届）英特尔杯大学生电子设计竞赛嵌入式 AI 专题赛

参赛队作品简介

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作品题目 (中英文对照)	慧语灵臂-基于云端/本地双模型的桌面机械臂抓放系统 DESKTOP ROBOTIC ARM PICK-AND-PLACE SYSTEM BASED ON SWITCHABLE CLOUD/LOCAL MODELS		
作品简介 (中英文对照，中文限500字以内)	“慧语灵臂”是一套基于 Intel Core Ultra 平台的本地化具身智能机械臂系统，面向桌面物体抓取、搬运、分类与堆叠任务。系统通过自然语言接收操作指令，利用云端或本地模型将指令解析为结构化任务，并在本地完成对象校验、动作白名单、关节限位及风险检查，避免模型直接控制执行机构。Gemini Pro Plus 相机采集 600×480 RGB 图像，视觉模块持续识别目标、夹爪和放置区域；机械臂采用“观察—规划—对准—抓取—验证—放置—再验证”的闭环流程，并结合 J6 位置反馈确认夹持与释放结果。系统支持 Windows、Ubuntu 和 ROS 2 环境，已实现总线舵机通信、关节控制、正运动学、数值雅可比、手动调试、急停及故障恢复等功能。设计目标为检测准确率大于 95%，夹爪与 TCP 定位误差小于 2 cm，视觉闭环收敛时间小于 3 s，单次任务耗时小于 1 min。作品具有云地模型切换、本地安全校验及抓取与放置双闭环等特色。 English Introduction: Huiyu Lingbi is a localized embodied-intelligence robotic arm system based on the Intel Core Ultra platform. It is designed for tabletop object picking, transportation, sorting, and stacking. Natural-language instructions are converted into structured tasks by a selectable cloud or local model, while object validation, action allowlists, joint limits, and risk checks are enforced locally to prevent the model from directly controlling actuators. A Gemini Pro Plus camera captures 600 × 480 RGB images, enabling continuous detection of targets, the gripper, and placement regions. The arm follows a closed-loop workflow of observation, planning, alignment, grasping, verification, placement, and final verification, with J6 position feedback used to confirm grasp and release results. The system supports Windows, Ubuntu, and ROS 2, and implements bus-servo communication, joint control, forward kinematics, a numerical Jacobian, manual debugging, emergency stops, and fault recovery. Its design targets include detection accuracy above 95%, gripper and TCP localization errors below 2 cm, visual-loop convergence within 3 s, and task completion within one minute.		

注： 1. 请使用小 4 号字（12 号字），单倍行距填写；
2. 每支参赛队限一名指导教师；

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